

# DATA ENGINEER — SOFTWARE & DATA PLATFORM

Company: NovaTech Solutions

Location: Remote — India

Employment Type: Full-time

Experience: 0–2 years

## ABOUT THE ROLE

NovaTech Solutions is looking for a Data Engineer to join its Software & Data Platform team. The role involves designing reliable data pipelines, developing backend services for data-intensive applications, working with relational and NoSQL databases, and supporting analytics and machine learning workflows.

The ideal candidate should have strong programming fundamentals, practical experience with Python and SQL, an understanding of data structures and algorithms, and the ability to build maintainable, production-oriented software.

## KEY RESPONSIBILITIES

- Design, develop, test, and maintain scalable data pipelines for collecting, transforming, validating, and delivering structured and semi-structured data.
- Build ETL and ELT workflows using Python, SQL, and modern data-processing tools.
- Develop REST APIs and backend services that expose reliable data and analytics functionality.
- Work with relational databases such as PostgreSQL and MySQL and understand database design, indexing, normalization, and query optimization.
- Work with NoSQL databases such as MongoDB when appropriate for application requirements.
- Clean, transform, validate, and analyze datasets to ensure data quality and consistency.
- Implement efficient data structures and algorithms for processing large datasets.
- Collaborate with software engineers, analysts, and data scientists to understand data requirements and deliver reusable data solutions.
- Write unit tests and integration tests for data-processing pipelines and backend services.
- Monitor pipeline reliability, investigate failures, and improve performance and maintainability.
- Use Git-based version control and follow clean-code, documentation, and code-review practices.
- Participate in technical discussions and contribute to architecture and engineering decisions.

## REQUIRED QUALIFICATIONS

- Bachelor's degree or equivalent education in Computer Science, Information Technology, Software Engineering, Data Engineering, or a related technical field.

- Strong proficiency in Python.
- Strong proficiency in SQL and relational database concepts.
- Good understanding of data structures and algorithms.
- Understanding of ETL/ELT concepts and data pipeline architecture.
- Experience developing REST APIs or backend services.
- Familiarity with Git and collaborative software development workflows.
- Understanding of software engineering principles including modular design, testing, debugging, and documentation.
- Strong problem-solving and analytical skills.

## **PREFERRED QUALIFICATIONS**

- Experience with FastAPI, Flask, or Django.
- Experience with PostgreSQL, MySQL, or other relational database systems.
- Experience with MongoDB or another NoSQL database.
- Familiarity with Pandas and NumPy.
- Exposure to Apache Spark, Apache Airflow, Kafka, or other modern data engineering technologies.
- Familiarity with Docker and containerized applications.
- Understanding of cloud platforms such as AWS, Microsoft Azure, or Google Cloud Platform.
- Exposure to data visualization or business intelligence tools.
- Knowledge of machine learning workflows and model-serving systems.
- Experience working with Redis or other caching technologies.
- Experience building production-quality projects or data-intensive applications.

## **TECHNICAL SKILLS**

Programming: Python, SQL

Backend: FastAPI, Flask, Django, REST APIs

Databases: PostgreSQL, MySQL, MongoDB, Redis

Data Processing: Pandas, NumPy, Apache Spark

Data Engineering: ETL, ELT, Data Pipelines, Data Validation

Development Tools: Git, Docker, Linux

Cloud: AWS, Azure, Google Cloud Platform

Testing: Pytest, Unit Testing, Integration Testing

Other: Data Structures, Algorithms, System Design, API Development

## **EDUCATION**

Bachelor's degree in Computer Science, Information Technology, Software Engineering, Data Engineering, or a related field.

## **WHAT WE VALUE**

- Strong engineering fundamentals and attention to detail.
- Ability to learn unfamiliar technologies quickly.
- Clear technical communication and documentation.
- Ability to reason about data quality, performance, and reliability.
- Practical project experience demonstrating software engineering ability.
- Ownership, curiosity, and a problem-solving mindset.
- Ability to work effectively in a remote engineering environment.

## **APPLICATION**

Candidates should submit a current resume highlighting relevant technical skills, projects, education, and practical software or data engineering experience.